

Private AI Server — Build Specification

Fully on-premise AI · no data leaves your servers · runs a 32B model (covers all planned AI features)

Prepared 2026-07-28 · Colocation-ready

Why 32B and not 96B: a 96B model needs far more GPU hardware. A **32B model** handles every planned feature — because the "intelligence" comes from your data + code (analytics, ranking, retrieval), while the model does the language (summarise, extract, draft, explain). You can upgrade the GPU later with no software changes.

Recommended build — 48 GB VRAM BEST VALUE

PART	SPECIFICATION
GPU	2× NVIDIA RTX 3090 24 GB (used) — 48 GB VRAM total. Best price/VRAM for local AI; NVLink-capable.
CPU	AMD Ryzen 9 (e.g. 5900X / 7900) — enough PCIe lanes for 2 GPUs at x8/x8 (fine for inference)
Motherboard	X570 / B650 with two PCIe x16 slots (runs x8/x8), good slot spacing
Memory	64 GB DDR4/DDR5
Storage	2 TB NVMe Gen4 SSD (OS + model weights + vector data)
Power supply	1000–1200 W, 80+ Gold or better (quality unit for 24/7)
Case & cooling	High-airflow case, extra fans; ensure clearance for 2 GPUs
Result	48 GB VRAM · runs a 32B model comfortably (70B 4-bit possible, tight)

Leaner start — validate first

PART	SPECIFICATION
GPU	1× NVIDIA RTX 3090 24 GB (used) — runs 14B–32B models
Rest of system	Same CPU / board / 32 GB RAM / 1 TB NVMe / 850 W PSU / case
Result	24 GB VRAM · prove the product, then add a 2nd 3090 for 48 GB

What this machine runs & enables

Models

- 32B instruct model (e.g. Qwen2.5-32B) — multilingual EN/DE/ES
- 14B model for high-volume cheap tasks
- Embeddings model (bge-m3) for semantic search
- Whisper for voice → text (field techs)

Features it powers

- Org analyst digests · smart dispatch suggestions
- Service-report auto-draft · NL → task / report / analytics
- Member coach · role-aware assistant · risk alerts
- Semantic search across jobs / assets / customers

Notes

- **Software (no licence cost):** Ubuntu + vLLM or Ollama (private, OpenAI-compatible endpoint) + pgvector on your existing PostgreSQL. Nothing leaves your network.
- **Upgrade path:** swap to bigger GPUs later (RTX 6000 Ada / A100 / H100) with **zero software changes** — the model sits behind an interface.
- **Longevity:** used consumer 3090s aren't ECC/datacenter-grade, but run reliably for years with good cooling.
- **Colocation:** a tower or 4U case in an EU datacenter keeps it GDPR-clean and always-on.

HBCField — Private AI initiative · Build spec (Step 1: Hardware). Next: Step 2 — pick the model.